STAFF REVIEW · H446 042 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A name scoped to a function or block

2. To modularise and reuse logic

3. A running total

4. Two keys produce the same location

I Do — source worked steps

1. Copy an integer value — Start x=5; CopyInt(x) changes only local n to 9. The caller x remains 5. This is pass-by-value.

2. Copy a reference; mutate one object — Start item→Box A(Value=5), alias=item. Mutate(item) copies the reference, then changes Box A.Value to 9. Both item and alias still refer to Box A.

3. Reassign the caller through ref — Replace(ref item) makes caller item refer to new Box B(Value=20). alias still refers to Box A(Value=9). Without ref, assigning a new Box to local p would leave item unchanged.

We Do — guided CHECK

1. Check your thinking
5.

2. Discuss, then respond
Does passing a reference-type object by value copy the object itself?
Check your thinking
No; the reference is copied.

You Do — source mark schemes

1. The method receives a copy of its value.

2. static void Inc(ref int value)

3. The former copies the reference, permitting object mutation; ref permits reassignment of the caller's variable.

Exit Ticket — source answer

1. False. Passing a reference-type variable normally passes a copy of the reference. Assigning a different object to that parameter does not reassign the caller's variable.